Sam Dillavou, Ph.D. dillavou@sas.upenn.edu

EDUCATION

Ph.D. in Physics, Harvard University	Ph.D. Thesis: *Hidden Dynamics of Static Friction*	2020
M.A. in Physics, Harvard University	Ph.D. Advisor: Shmuel M Rubinstein	2016
B.A. in Physics, Cornell University	Undergraduate Research Advisor: Itai Cohen	2012

RESEARCH EXPERIENCE

ARIA R&D Creator (Independent Researcher) 2025–2027
Topics: Learning Metamaterials | Distributed & Analog Machine Learning

Postdoctoral Fellow, University of Pennsylvania with Douglas J Durian & Andrea J Liu 2020–2025
Topics: Emergent Physical Learning | Granular Flows | ML in Experimental Science

Visiting Researcher, EPFL, Switzerland with John M Kolinski 2018
Topic: Developed ultrafast (≥ MHz) imaging technique for any camera

FUNDING AND AWARDS

Grants & Fellowships

ARIA Opportunity Seed Grant	£500k (~$650k) 2025–2027
AI Fellow, U Pennsylvania	$8k/year, 2025
Center for Soft and Living Matter Fellow, U Pennsylvania	$5k/year, 2025
Data Science Postdoctoral Fellow, U Pennsylvania	$5k/year, 2022–2024
Smith Family Fellowship, Harvard U	$90k, 2015–16
Purcell Fellowship, Harvard U	$90k, 2014–15

Research & Teaching Recognition

1st place GSNP Postdoctoral Presentation Award	APS March Meeting 2024
1st place Meeting-Wide Postdoctoral Poster Competition	APS March Meeting 2023
Herbert B. Callen Memorial Prize	University of Pennsylvania 2023
2nd place, title: Learning Networks on the Radio	MRSEC National Science Slam 2022
Editor's Suggestion, Physical Review Applied	Dillavou et al. 2022
Rising Star in Soft and Biological Matter	University of Chicago 2021
Editor's Suggestion, Physical Review Letters	Dillavou & Rubinstein 2018
Bok Center Certificate of Teaching Excellence	Harvard University Spring 2018

PUBLICATIONS

▪ equal contribution • work performed as an undergraduate

Lead Author

[1] **S Dillavou**, JW Rocks, JF Wycoff•, AJ Liu, DJ Durian. arXiv 2511.17825
Analog Physical Systems Can Exhibit Double Descent

[2] **S Dillavou**, S Mishra. arXiv 2511.17789
Physical Reinforcement Learning

[3] **S Dillavou**▪, M Guzman▪, AJ Liu, DJ Durian. arXiv 2505.22887
Understanding and Embracing Imperfection in Physical Learning Networks

[4] JM Hanlan▪, **S Dillavou**▪, AJ Liu, DJ Durian. Soft Matter, 2025
Cornerstones are the Key Stones: Using Interpretable Machine Learning to Probe the Clogging Process in 2D Granular Hoppers

[5] D Hathcock▪, **S Dillavou**▪, JM Hanlan, DJ Durian, Y Tu. Physical Review E, 2025
Stochastic dynamics of granular hopper flows: a configurational mode controls the stability of clogs

[6] (Perspective) **S Dillavou**. Physics Magazine, 2024
Harnessing Machine Learning to Guide Scientific Understanding

[7] **S Dillavou**, B Beyer•, M Stern, AJ Liu, MZ Miskin▪, DJ Durian▪. PNAS, 2024
Machine Learning Without a Processor: Emergent Learning in a Nonlinear Analog Network

[8] **S Dillavou**, JM Hanlan, AT Chieco, H Xiao, S Fulco, K Turner, DJ Durian. Scientific Reports, 2024
Bellybutton: Accessible & Customizable Deep-Learning Image Segmentation

[9] **S Dillavou**, Y Bar-Sinai, MP Brenner, and SM Rubinstein. Physical Review E, 2022
Beyond Quality and Quantity: Spatial Distribution of Contact Encodes Frictional Strength

[10] **S Dillavou**, M Stern, AJ Liu, DJ Durian. Editor's Choice, Physical Review Applied, 2022
Demonstration of Decentralized, Physics-Driven Learning

[11] **S Dillavou** and SM Rubinstein. Physical Review Letters, 2020
Shear Controls Frictional Aging by Erasing Memory

[12] **S Dillavou**, SM Rubinstein, and JM Kolinski. Optics Express, 2019
The Virtual Frame Technique: Ultrafast Imaging With Any Camera

[13] **S Dillavou** and SM Rubinstein. Editor's Choice, Physical Review Letters, 2018
Nonmonotonic Aging and Memory in a Frictional Interface

Additional

[14] KA Murphy, **S Dillavou**, DS Bassett. Transactions on ML Research, 2025
Comparing the information content of probabilistic representation spaces

[15] AJ Gerra, CC Jones, **S Dillavou**, JM Hanlan, J Radzio, PE Arratia, DJ Durian. Physical Review Applied, 2024
The Equation of Motion for Taut-Line Buzzers

[16] T Martin, **S Dillavou**. J of Cultural Cognitive Science, 2024
Calculations Without Math: "Smart instruments" & the transposition of complex shapes in the wooden boat workshop

[17] M Stern, **S Dillavou**, D Jayaraman, DJ Durian, AJ Liu. APL Machine Learning, 2024
Training self-learning circuits for power-efficient solutions

[18] W Steinhardt, **S Dillavou**, M Agajanian, SM Rubinstein, EE Brodsky. Geophysical Research Letters, 2023
Seismological Stress Drops for Confined Ruptures Are Invariant to Normal Stress

[19] A Srivastava … **S Dillavou** … Z Wu (100s of authors). Transactions on ML Research, 2023
Beyond the Imitation Game: Quantifying & extrapolating the capabilities of language models

[20] M Pasquet, … AT Chieco, **S Dillavou**, JM Hanlan, DJ Durian, E Rio, A Salonen, D Langevin. Comptes Rendus Mécanique, 2023
Aqueous foams in microgravity, measuring bubble sizes

[21] SCL Durian, **S Dillavou**, K Markin, A Portales, BOT Maldonado, WTM Irvine, PE Arratia, DJ Durian. Physics of Fluids, 2022
Spatters and Spills: Spreading Dynamics for Partially Wetting Droplets

[22] JF Wycoff, **S Dillavou**, M Stern, AJ Liu, DJ Durian. Journal of Chemical Physics, 2022
Desynchronous Learning in a Physics-Driven Learning Network

[23] M Stern, **S Dillavou**, MZ Miskin, DJ Durian, AJ Liu. Physical Review Research, 2022
Physical Learning Beyond the Quasistatic Limit

[24] S Zheng, **S Dillavou**, JM Kolinski. Soft Matter, 2021
Air Mediates the Impact of a Compliant Hemisphere on a Rigid Smooth Surface

[25] T Pilvelait, **S Dillavou**, and SM Rubinstein. Physical Review Research, 2020
Influences of Microcontact Shape on the State of a Frictional Interface

[26] JL Silverberg, **S Dillavou**, L Bonassar, and I Cohen. J Orthopaedic Res, 2013
Anatomic Variation of Depth-Dependent Mechanical Properties in Neonatal Bovine Articular Cartilage

Conference Workshop Proceedings

[a] **S Dillavou**, B Beyer, M Stern, MZ Miskin, AJ Liu, DJ Durian. NeurIPS ML with New Compute Paradigms Workshop, 2023
Nonlinear Classification Without a Processor

[b] M Stern, **S Dillavou**, D Jayaraman, DJ Durian, AJ Liu. NeurIPS ML with New Compute Paradigms Workshop, 2023
Contrastive power-efficient physical learning in resistor networks

[c] **S Dillavou**, B Beyer, M Stern, MZ Miskin, AJ Liu, DJ Durian. Proceedings SPIE, AI & Optical Data Sciences IV, 2023
Circuits that train themselves: decentralized, physics-driven learning

[d] M Stern, **S Dillavou**, MZ Miskin, DJ Durian, AJ Liu. NeurIPS ML & the Physical Sciences Workshop, 2021
Out of Equilibrium Learning Dynamics in Physical Allosteric Resistor Networks

Patents

[P] **S Dillavou**, DJ Durian, AJ Liu, M Stern, MZ Miskin. US Patent 12462202 B2, 2025
Coupled Networks for Physics-Based Machine Learning

Software Packages Authored

Bellybutton – a Python deep learning package for image segmentation, designed for researchers with no coding required. Download available here. Associated publication: [8]

TEACHING AND MENTORING EXPERIENCE

Research Mentee	Career Stage, University	Coauthored Publications	Year(s)
Jacob F Wycoff	Postgrad	[1][22]	2025–26
Alex Roseman	Undergraduate, Yale U		2024
Juan Mendez	Undergraduate, Williams		2024
Evan Stocker	Undergraduate, Pennsylvania State U		2024
Benjamin D Beyer	Undergraduate, U of Pennsylvania	[7][a][c]	2021–24
Jesse M Hanlan	Ph.D. Student, U of Pennsylvania	[4][5][8][15][20]	2020–24
Josue D Ruiz	Undergraduate, U of Pennsylvania		2022–23
Jacob F Wycoff	Undergraduate, U of Pennsylvania	[1][22]	2021–23
Courtney C Jones	Undergraduate, U Maryland	[15]	2022
Alex Gerra	Undergraduate, Moravian U	[15]	2022
Kwame Markin	Undergraduate, Swarthmore	[21]	2021
Adrian Portales	Undergraduate, U Texas Rio Grande	[21]	2021
Sylvia CL Durian	Undergraduate, U Chicago	[21]	2021
Mary Agajanian	Undergraduate, Harvard U	[18]	2019–21
Tom Pilvelait	Undergraduate, Harvard U	[25]	2018–20
Vincent Stin	Undergraduate, ESPCI, Paris		2019
Aijie Xu	Ph.D. Student, Tsinghua U		2017–18
Evgeni Shirman	Undergraduate, Hebrew U, Jerusalem		2016

Teaching

Substitute Lecturer: Analytical Mechanics, Douglas Durian – 30 Undergrads — U Penn, Spring 2024

TA: Introduction to Fluid Mechanics, Shmuel Rubinstein – 60 Undergrads — Harvard U, Spring 2018
  Develop new materials, in-class demos, grade assignments, supervise labs and final projects
  Received Bok Center Certificate of Teaching Excellence

TA: Introduction to Soft Matter, Shmuel Rubinstein – 20 Grad Students — Harvard U, Fall 2015
  Write problem sets, develop new materials, teaching section, grading

Short Courses / Workshops / Tutoring

Improving Presentation & Discussion Through Improvisation – 15 Grad. — Harvard U, Winter 2019

Intro to Long-Form Improvisation – 15 Graduate Students — Harvard U, Winter 2016
  Developed and taught custom curricula for both mini-courses

Improvisational Theater Workshops — Harvard, Tufts, Yale, Cornell, Deloitte, 2012–Present
  Designed & taught for 5–50 participants, 6th grade to grad, business professionals, faculty.

High school/college math, physics, and SAT tutoring; 100s of hours, 30+ students — 2010–Present

Pedagogical Training

Teaching and Communicating Physics — Harvard U, Spring 2015

PRESENTATIONS AND PRESS

[some titles truncated for space]

Invited Talks

Analog Hardware That Trains Itself	Frontiers of Neuromorphic Computing II, Max Planck, 2026
Emergent Learning in Analog Electronic Networks	(AI)² Research Talk Series, Princeton U, 2025
Self-Learning Networks in the Lab	Beyond von Neumann Computing Wkshp, Aspen, CO, 2025
M.L. as an Experimental Guide	PREM20: Self-Driving Science: AI & AE, San Juan, PR, 2025
Emergent Learning in Analog Electronic Networks	Physics/Math Special Seminar, NYU, 2025
The Effects of Physicality on Emergent Learning	APS March Meeting, Anaheim, CA, 2025
How to Construct a Learning 'Material' (Tutorial)	APS March Meeting, Anaheim, CA, 2025
Emergent Physical Learning	Applied Physics Special Seminar, Cornell U, 2024
Emergent Physical Learning	Tufts Condensed Matter Seminar, 2024
Emergent Machine Learning	Inaugural klogW Future Series Seminar, Virtual, 2024
The Metamaterial that Trains Itself	SIAM: Mathematical Aspects of Mat Sci, Pittsburgh, PA, 2024
Emergent Learning in Electronic Networks	Physics Dept Special Seminar, U Chicago, 2024
Supervised Learning as an Emergent Property	Argonne Nat'l Lab Applied AI Series, Virtual, 2023
Evolution of a Learning Material	9th IDMRCS, Chiba, Japan, 2023
Decentralized, Physics-Driven Learning	SPIE Photonics West, San Francisco, CA, 2023
A Physics-Driven Learning Network	Alternative Computing Group Seminar, NIST, 2023
Demonstration of Decentralized, Physics-Driven Learning	Phys Rev Journal Club, Virtual, 2022
Hijacking Physics to Learn for Us	Weekly Seminar, Google Brain, 2022

Using Physics to Learn without a Processor — APS March Meeting, Chicago, IL, 2022
Decentralized Physics-Driven Learning — Physics Seminar, Bucknell U, 2021
Hidden Dynamics of Static Friction — Applied Math Seminar, NYU, 2020
Hidden Dynamics of Static Friction — Soft Matter Coffee Hour, Princeton U, 2020
Hidden Dynamics of Static Friction — Soft Matter Theory Group (U Pennsylvania), Virtual, 2020
Static Friction: Aging and Memory — Geomechanics Seminar, Pennsylvania State U, 2018
Memory in Solid-Solid Interfaces — Mechanical Engineering Seminar, EPFL, 2018

Selected Press

On Physical & Emergent Learning

Machines That Learn Like Physicists Think — APS Research Highlights, 2026
How a simple circuit could offer an alternative to energy-intensive GPUs — MIT Tech Review, 2024
A first, physical system to learn nonlinear tasks without a traditional processor — Penn Today, 2024
Training neural networks using physical equations of motion — PNAS Physics Commentary, 2024
How to make the universe think for us — Quanta Magazine, 2022
Simple electrical circuit learns on its own – with no help from a computer — Science News, 2022
Programming matter to a computer's job — American Physical Society News, 2021

On the Virtual Frame Technique

Imaging technique lets ordinary cameras capture high-speed images of crack formation — Phys.org, 2019
How to mod a smartphone camera so it shoots a million frames per second — MIT Tech Review, 2018

On Memory in Frictional Interfaces

Friction Remembers Its Origins — American Physical Society Physics Focus, 2018
Friction Remembers Its Past — Physics Today, 2018

Contributed Talks

Local Learning Rules in Analog Hardware — Computing with Physical Systems, 2026
Studying Granular Clogging with ML as an Experimental Guide — NE Granular Mat, Holy Cross, 2024
Emergent Learning Via Sequential Error Mode Reduction — APS March Mtg, Minneapolis, MN, 2024
    1st Prize Statistical & Nonlinear Physics Postdoctoral Speaker Award
Transistor-Based Self-Learning Networks — APS March Meeting, Las Vegas, NV, 2023
A Physics-Driven Self-Learning Transistor Network — Coherent Network Comp., Stanford U, 2022
Contact Distribution Encodes Frictional Strength — APS March Meeting, Chicago, IL, 2022
Decentralized Physics-Driven Learning — Rising Stars in Soft & Biological Matter, U Chicago, 2021
Building a Physical Learning Network — APS March Meeting, Virtual, 2021
Memory in Solid-Solid Interfaces — APS March Meeting, Boston, MA, 2019
Hidden Dynamics of Static Contact and Static Friction — Dynamics Days, Northwestern U, 2019
Extreme Mechanics of Elastomer Impact — NORA & BASF Collab. Days, U Mass Amherst, 2019
Two Solids Make a Glass: Memory in Solid-Solid Interfaces — APS March Mtg, Los Angeles, CA, 2018
Elastomer Wear: The NBA's Shoe Problem — NORA & BASF Collab. Days, U Mass Amherst, 2017
Memory in the Frictional Interface — Physics Dept Mini-Symposium, Weizmann Inst, 2017

Posters/Rapid Talks

Nonlinear Classification Without a Processor — Ctr for Soft and Living Matter Kickoff, UPenn, 2024
Self-Learning Electronic Networks — Mid-Atlantic Soft Matter Workshop, Georgetown U, 2024
Nonlinear Classification Without a Processor — Computing with Physical Systems, Aspen, CO, 2024
A Physics-Driven Self-Learning Transistor Network — APS March Meeting, Las Vegas, NV, 2023
  1st Prize in the APS March Meeting Postdoctoral Poster Competition
Physical Learning Machines — Cracking the Glass Problem Simons Mtg, New York, NY, 2022
Building a Physical Learning Network — Northeast Complex Fluids Workshop, Virtual, 2021
Tabletop Nucleation — Southern California Earthquake Center Annual Mtg, Palm Springs, CA, 2019
The Hidden Dynamics of Static Friction — Gordon Conf: Soft Matter Phys., New London, NH, 2019
The Virtual Frame Technique — 77th New England Complex Fluids, Harvard U, 2018
Memory in the Frictional Interface — 73rd New England Complex Fluids, Harvard U, 2018
Memory in the Frictional Interface — Jay (Fineberg) Fest, Hebrew U in Jerusalem, 2017
Beyond Rate and State: Frictional Memory — Inst. for Study of the Continents Conf, Cornell U, 2017
Wear in Basketball Shoes — NORA & BASF Challenges, U Mass Amherst, 2017
Visualizing Frictional Interfaces — 69th New England Complex Fluids, Harvard U, 2016
Visualizing Frictional Interfaces — 67th New England Complex Fluids, MIT, 2016
Loading History of Frictional Interfaces — Phys + Mech of Soft Complex Mat, Cargese, France, 2016
Loading History of Frictional Interfaces — Gordon Conference: Tribology, Lewiston, ME, 2016
Visualizing Growth of a Multi-contact Interface — 65th NE Complex Fluids, Harvard U, 2015
Aging of Multi-Contact Interfaces — Soft Matter: Friction, Rheology, Tribology, U Florida, 2015

PROFESSIONAL SERVICE

Journal Referee

Science, Nature [Communications, Electronics], Soft Matter, Physical Review [B, E, Applied, Letters], US Geological Survey Internal, J of Geophysical Research – Solid Earth, NeurIPS Workshops

Outreach

Design and teach U Penn Data Driven Discovery Initiative ML Workshop — 2023, 2024, 2025
Philly Materials Day (K–12), design, construct and demo trainable elastic material — 2024
Design and teach U Penn REU Machine Learning (ML) Workshop — 2022, 2023, 2024
DEEPenn STEM (see below), volunteer, mentor, presenter — 2023
"Friction: The surprising unsolved science behind earthquakes and tire treads" — Science Café, Wilmington, Delaware, 2023
Planning committee, volunteer, presenter for the first annual DEEPenn STEM: weekend-long STEM PhD prep/info workshop for ~45 URM college students from around the country — 2022
MRSEC National Science Slam: Learning Networks on the Radio — 2022
Science in the News Writer, Harvard U — 2016–17
Splash at Yale Instructor, grades 7–9 and 10–12, Yale U — 2016, 2017

Professional Membership

American Physical Society — 2016–Present
APS March Meeting session organizer, chair, and sorter — 2023, 2024

Miscellaneous

Part of a collaboration developing a 3D Printer-as-Ventilator during COVID-19 outbreak — 2020